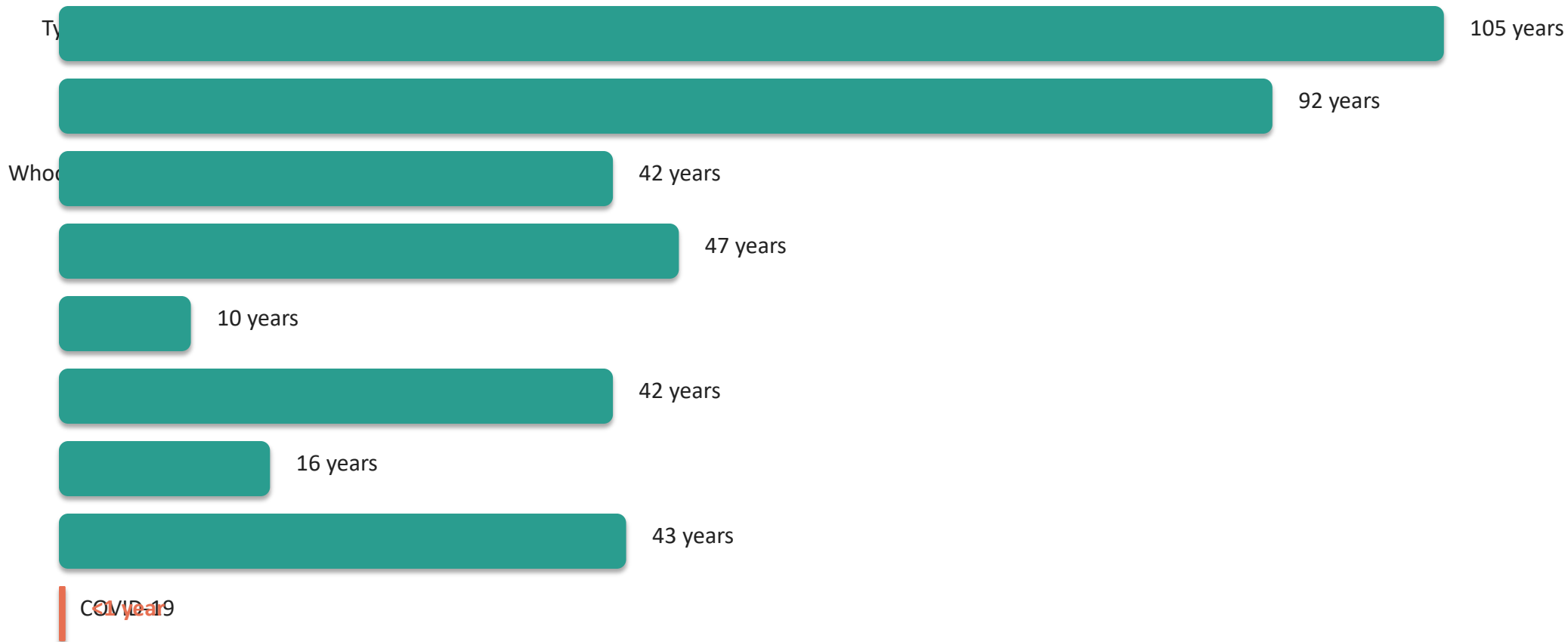


The Power of Vaccination: From Smallpox Eradication to COVID-19 in Record Time

A data-driven overview of vaccine development, disease elimination, and remaining global coverage gaps (1796–2020)

Source: Our World in Data • Audience: Public health professionals & policy analysts

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating



Data: Pathogen identification to vaccine licensure. Sources: Roush & Murphy (2007); Our World in Data.

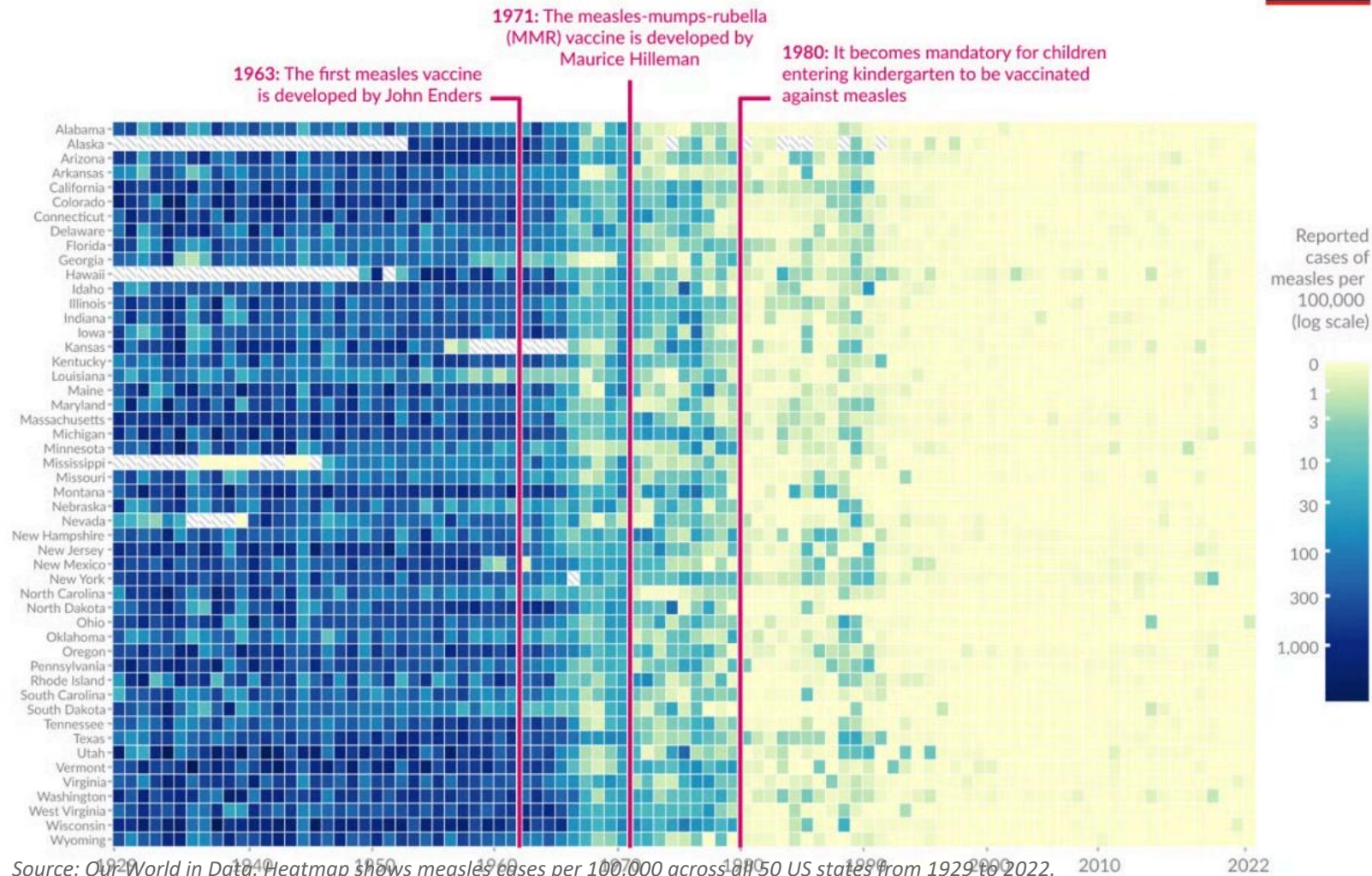
100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

Disease	Pre-Vaccine Cases (per million/yr)	Post-Vaccine Cases (per million/yr)	Case Reduction	Pre-Vaccine Deaths (per million/yr)	Death Reduction
Diphtheria	158	0	100%	13.7	100%
Measles	3,044	0.2	99.99%	2.5	100%
Smallpox	250	0	100%	2.9	100%
Acute polio	141	0	100%	10	100%
Pertussis	1,534	52	96.6%	30.8	99.7%
Rubella	242	0.04	99.98%	0.09	100%
Hepatitis A	465	51	89%	0.5	88.7%
Acute hepatitis B	273	44	83.9%	1	83.6%
Pneumococcal	233	139	40.5%	24	31.3%

Data: Roush & Murphy (2007). Case and death rates per million population per year in the United States.

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

Vaccines reduced measles cases across US states



1963

First measles vaccine developed by John Enders

1971

MMR (measles-mumps-rubella) vaccine developed by Maurice Hilleman

1980

Vaccination became mandatory for children entering kindergarten

Before 1963, nearly every state experienced high measles incidence (100–1,000+ cases per 100,000). After the 1980 mandate, cases dropped to near zero across all states, with only sporadic outbreaks.

Source: Our World in Data. Heatmap shows measles cases per 100,000 across all 50 US states from 1929 to 2022.

Data source: Project Tycho (2018); Centers for Disease Control and Prevention (1959–2022)

OurWorldinData.org — Research and data to make progress against the world's largest problems

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Smallpox: The Only Human Disease Eradicated Through Vaccination

1959

WHO passes resolution to eradicate smallpox

1967

Intensified eradication campaign launched

1977

Last natural case occurs in Somalia

1980

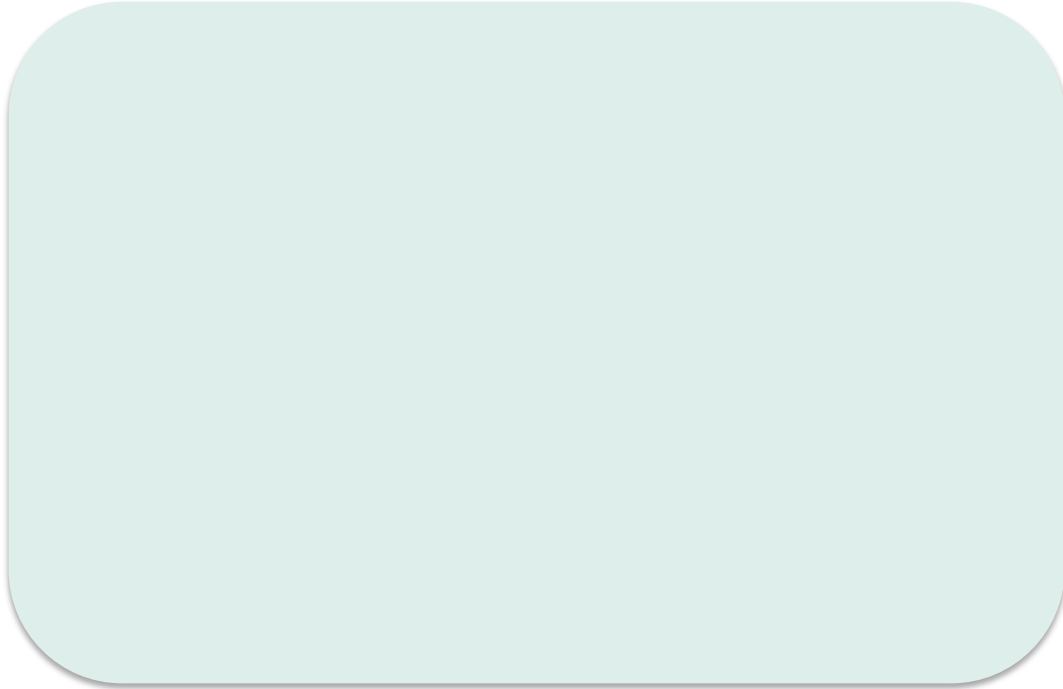
Smallpox declared eradicated worldwide

Why was smallpox uniquely suited for eradication?

- The virus infected only humans (no animal reservoir)
- Symptoms were visually distinguishable, enabling targeted containment
- The vaccine was cheap, stable, and highly effective

Smallpox remains the only human disease to be fully eliminated through vaccination — a milestone certified by the WHO in 1980.

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable



High vaccination coverage creates a community shield



Vaccinated (immune)



Vulnerable (cannot be vaccinated)

When enough people are vaccinated, pathogens cannot spread effectively.

This shields those who cannot be vaccinated:

- Newborns
- Immunocompromised individuals
- Cancer patients on chemotherapy

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

Pneumococcal Disease

40.5% case reduction
31.3% death reduction

Lowest impact among major vaccine-preventable diseases in the US. Highlights need for improved vaccines and broader coverage.

Hepatitis A & B

89% and 83.9% case reduction

Incomplete control remains. Access to vaccination and targeted outreach for at-risk populations are critical.

Global Access & Trust

Supply shortages & hesitancy

Not everyone has access to vaccines. Poor scientific understanding reduces trust and uptake, putting children and communities at risk.

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

- 1 Vaccines achieved 100% elimination of cases and deaths for diphtheria, smallpox, and acute polio in the US.
- 2 Vaccine development timelines have collapsed from over 100 years (typhoid) to under 1 year (COVID-19), driven by mRNA technology and genomic tools.
- 3 Measles cases dropped to near zero across all 50 US states following the 1980 kindergarten vaccination mandate.
- 4 Smallpox remains the only human disease eradicated through vaccination, certified in 1980 after a global campaign.
- 5 Gaps persist for pneumococcal disease (40.5% case reduction), hepatitis A (89%), and hepatitis B (83.9%), highlighting the need for improved access, coverage, and newer vaccines.

Call to Action for Global Health Systems

Strengthen immunization infrastructure, close equity gaps, invest in next-generation platforms, and build public trust through transparent science.